

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS: Total Marks: 20

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser.** Your answer should clearly describe the purpose and role of each concept in web applications.

2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. ***Signature of the student:***

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are	Minor technical errors	Some incorrect or vague	Major technical errors

		technically correct and precise		explanations	
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

① XML - Extensible Markup Language

XML is a Markup Language used to store, organize, and transport data in a structured format.

- It allows developers to create their own tags.
- It separates data from its presentation.
- It is commonly used for exchanging data between different applications.

JSP - Java Server Pages

JSP is a Java-based server-side technology used to create dynamic web pages.

- JSP allows HTML pages to contain Java-based server-side logic.
- The JSP page is processed by a web server, which generates HTML and sends it to the browser.
- It is commonly used for displaying database information dynamically.

MVC Architecture - Model view Controller

MVC is a Software architecture that divides an application into three Components.

- Model : Handles data and business Logic
- View : Handles the user interface and presentation,
- Controller : Receives user requests and Controls the flow b/w model and view.

XSLT - Extensible stylesheet Language Transformations

XSLT is a Language used to transform XML documents into other formats such as HTML, XML or text.

For example, XML containing student records can be transformed into an HTML table using XSLT.

XML Schema

XML Schema defines the structure, rules, and data types of an XML document.

It can specify:

- Which elements are allowed
- The order of elements
- Data types such as string, integer or date.

JSTL - Jsp Standard Tag Library

JSTL is a Collection of Standard tags used in Jsp pages to perform Common tasks without writing large amounts of Java Code.

It provides tags for:

- Loops
- Conditions
- Formatting
- Database Operations
- XML Processing

PHP - HyperText Preprocessor

PHP is a server-side scripting Language widely used for developing dynamic and interactive websites.

- PHP Code runs on the web server.
- It can Process forms and user requests.
- It can Connect to databases.
- It can generate HTML dynamically.

Database Connectivity

Database Connectivity is the process of allowing a web application to communicate with a database.

It allows applications to:

- Connect to database.
- Insert new data.
- Retrieve data.
- Update existing data.
- Delete data.

Parser

A parser is a software component that reads and analyzes structured data according to its rules and converts it into a form that a program can understand.

For example, an XML parser reads an XML document and checks whether it follows XML syntax. It can then provide the data to an application.

- ② In a modern web application, technologies such as XML, Jsp, PHP, MVC architecture, database Connectivity and parson work together to ensure efficient data processing, Communication, and application management.

1. XML (Extensible Markup Language)

XML is used to store and transfer data between different systems. It organizes information using custom tags in a hierarchical structure.

Example

```
<student>
  <ID>101</ID>
  <Name>Harshath</Name>
  <Department>IT</Department>
</student>
```

- XML documents can be validated using DTD or XML Schema.
- Validation ensures that the XML follows predefined rules and maintains data consistency.

2. Jsp (Java Server Pages) and PHP

- Run on a Java web server such as Apache Tomcat.
- processes user requests and interacts with databases.

Ex:-

```
<? = "Welcome user" %>
```

PHP

- Executes on a php-enabled web server.
- Handles form processing, authentication and database operations

Ex:-

```
<?php  
echo "welcome user";  
?>
```

3. MVC (Model - View - Controller) Architecture

MVC separates an application into three components.

Model

- Handles data and business logic
- Communicates with the database

View

- Displays information to user.
- Usually consists of HTML, CSS, JSP or PHP pages.

Controller

- Receives user requests.
- Processes requests and coordinates b/w model and view.

4. Database Connectivity

Database Connectivity enables web application to store and retrieve data.

JDBC is used :

```
Connection con = DriverManager.getConnection (url, user, password);
```

In PHP MySQL or PDO is commonly used.

```
$conn = mysqli_connect ("localhost", "root", "", "College");
```

Database Connectivity allows application to:

- store user information
- Retrieve records.

5. Parsers

Parsers read and process structured data such as XML.

Types of XML parsers

1. DOM parser

- Loads the entire XML document into memory.
- Suitable for smaller files.

2. SAX parser

- Reads XML sequentially.
- Faster and Memory-efficient for large files.